

CO-1

ASSESSMENT TOOL 3

CASE STUDY

NAME: Y.SAI PRASANTH VARMA

REGISTRATION NUMBER: 192411191

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4308

Scenario: Web Development Standards Implementation

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`

D. <note to="User"></note>

Snippet 5: Web Communication Protocols

A. FTP

B. HTTP

C. SMTP

D. SNMP

Source Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Web Development Standards Implementation</title>
```

```
<style>
```

```
  body{
```

```
    font-family: Arial, sans-serif;
```

```
    margin:20px;
```

```
    background:#f4f4f4;
```

```
  }
```

```
  header{
```

```
    background:#0077cc;
```

```
    color:white;
```

```
    padding:15px;
```

```
    text-align:center;
}
```

```
nav{
    background:#333;
    padding:10px;
    text-align:center;
}
```

```
nav a{
    color:white;
    text-decoration:none;
    margin:15px;
    font-weight:bold;
}
```

```
section{
    background:white;
    margin-top:20px;
    padding:20px;
    border-radius:10px;
}
```

```
input{
    width:100%;
    padding:10px;
    margin:8px 0;
```

```
}
```

```
button{  
    padding:10px 20px;  
    background:green;  
    color:white;  
    border:none;  
    cursor:pointer;  
}
```

```
video{  
    width:500px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Startup Web Application</h1>
```

```
</header>
```

```
<!-- Semantic Navigation -->
```

```
<nav>
```

```
<a href="#">Home</a>
```

```
<a href="#">About</a>
```

```
<a href="#">Services</a>
```

```
<a href="#">Contact</a>

</nav>

<!-- Video Section -->

<section>

<h2>Promotional Video</h2>

<video controls>

  <source src="video.mp4" type="video/mp4">

  Your browser does not support the video tag.

</video>

</section>

<!-- Registration Form -->

<section>

<h2>User Registration</h2>

<form action="register.php" method="POST">

<label>Name</label>

<input type="text" name="name" required>

<label>Email</label>

<input type="email" name="email" required>
```

<label>Password</label>

<input type="password" name="password" required>

<button type="submit">Register</button>

</form>

</section>

</body>

</html>

Questions

1. Select the correct HTML tag for video embedding and justify your answer.

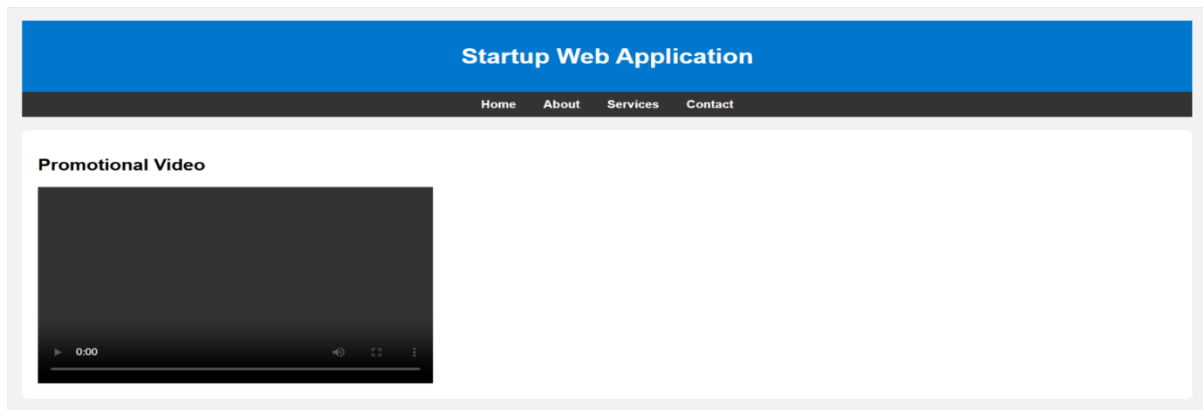
Correct Answer: B. <video src="video.mp4" controls></video>

Justification:

- The <video> tag is the standard HTML5 element used to embed videos in web pages.
- The controls attribute provides play, pause, volume, and fullscreen controls for users.
- It is supported by all modern web browsers.

Why the other options are incorrect:

- **A.** <media> – Not a valid HTML element.
- **C.** <movie> – Not an HTML tag.
- **D.** <vid> – Not a valid HTML tag.



2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

Explanation:

- The POST method is used to securely send form data to the server.
- Data is sent in the request body instead of the URL, making it suitable for sensitive information such as usernames, passwords, and registration details.
- It supports large amounts of data and does not expose user information in the browser address bar.

Why the other options are incorrect:

- **A. GET** – Sends data in the URL; not suitable for sensitive information.
- **C. PUT** – Used to update existing resources.
- **D. DELETE** – Used to remove resources from the server.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. <nav>

Justification:

- The <nav> element is the semantic HTML tag specifically designed for navigation links.
- It improves website accessibility and helps search engines understand the page structure.
- It makes the HTML code more meaningful and easier to maintain.

Why the other options are incorrect:

- **A.** <div> – Generic container with no semantic meaning.
- **C.** <section> – Used to group related content, not specifically for navigation.
- **D.** <header> – Represents introductory content and may contain navigation, but it is not dedicated to navigation.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: B. <note><to>User</to><from>Admin</from></note>

Explanation:

- All XML elements are properly opened and closed.
- Tags are correctly nested.
- The document is well-formed.

Errors in other options:

- **A.** <note><to>User</to><from>Admin</from>
 - Missing the closing </note> tag.
- **C.** <note><to>User</from></note>
 - Incorrect closing tag (</from> instead of </to>).
- **D.** <note to="User"></note>
 - Missing the closing angle bracket (>) in the end tag, resulting in invalid XML syntax.

Correct XML Syntax

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP

Justification:

- HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers.
- It is responsible for transferring web pages, images, videos, and other web resources.

- Modern websites commonly use HTTPS, which is the secure version of HTTP.

Why the other options are incorrect:

- **A.** FTP – Used for file transfer.
- **C.** SMTP – Used for sending email.
- **D.** SNMP – Used for network monitoring and management.

Web Communication Protocol

This webpage is intended to be accessed using the HTTP/HTTPS protocol, which enables communication between the web browser and the web server.